

Ayush Mitesh Nimkar

 Ayush-Nimkar |  Ayush Nimkar |  ayushmn29@gmail.com |  +91 9426242597

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT) Surat, Gujarat
B.Tech in Electronics & Communication Engineering (3rd Year) 2024 – Present
Cumulative GPA: 7.5/10.0 | 4th Semester SGPA: 8.0/10.0

Indian School Al Wadi Al Kabir Muscat, Oman
Class 12 Board Examination 2023 – 24
Percentage: 91%
Class 10 Board Examination 2021 – 22
Percentage: 92.6%

EXPERIENCE

Literary Affairs Committee (LAC), SVNIT Surat, Gujarat
Quiz Head Aug 2024 – Present

- Spearheaded the planning and execution of Litfest 2025, leading a team of student volunteers to manage logistics, promotions, and scheduling for university-wide literary events. Demonstrated strong cross-functional teamwork and leadership by coordinating with various campus departments to streamline event operations, and drive overall student engagement.

PROJECTS

UART Protocol IP Core | *Verilog, FSM Design, Digital Logic* 2026

- Architected and implemented a Universal Asynchronous Receiver-Transmitter (UART) protocol IP in Verilog, enabling reliable serial communication between digital peripherals.
- Developed custom Moore Finite State Machines (FSMs) for both Transmitter (TX) and Receiver (RX) datapaths, accurately handling start-bit detection, data framing, and stop-bit generation.

Hardware Logic & Controller Designs | *Verilog, RTL Simulation* 2026

- Programmed and simulated a suite of hardware control systems using Verilog, including a Water Reservoir Controller, Traffic Light Controller, and an Elevator Controller. Implemented complex cellular automaton logic by designing a hardware-based Conway's Game of Life.

Autonomous Line Follower Robot | *Arduino, Hardware Design* 2025
Mindbend Tech Event, SVNIT

- Engineered an autonomous line-following robot for MindBend2k25, integrating sensors and microcontroller-based logic for real-time path tracking.

Solar-Powered Obstacle Avoiding Robot | *Arduino, Renewable Energy* 2022

- Constructed a robotic hardware prototype leveraging an Arduino micro-controller and ultrasonic sensors for dynamic obstacle detection and automated collision avoidance, integrating a solar panel power system to create a self-sustaining platform.

SKILLS & CERTIFICATIONS

- **Hardware & Programming Languages:** Verilog, SystemVerilog, C, Python, Arduino, CMOS technology.
- **Data Science with Python Certification from Great Learning.**